COURSE

AQA GCSE Business (8132)

Financial calculations: ARR

3.6 Finance · 3.6.3 Financial terms and calculations: average rate of return

Revision for 3.6 Finance: each firm works it out on its own deals.

Class **AQA GCSE Business (8132)** Topic **Financial calculations: ARR**

THE LESSON

Setup	2 min

Round 1 — Decisions 3 min · Deals 5 min · Topic tasks 4 min

Firm decisions	**Deals**	TASK 1 **Lifetime profit**	TASK 2 **What ARR misses**

Round 2 — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min

Firm decisions	**Deals**	TASK 3 **ARR steps in order**	TASK 4 **Valeting bay ARR**	**B2B deal**

Round 3 — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min

Firm decisions	**Deals**	TASK 5 **ARR below loan rate**	TASK 6 **Better option ARR**	**B2B deal**

Final questions	3 min

45-minute setting shown. The 30-minute setting has one topic task per round (Tasks 3, 5 and 6) and one B2B deal. 60 and 75 add more deals.

Each step

Firm decisions	Each role makes its decision on its own page.
Deals	Each pupil trades with a classmate in another firm.
Topic task	The firm answers together. Everyone presses Agree.
B2B deal	One firm sells a lot of cars to another firm.
Final questions	Each pupil answers six questions.

Firm decisions

CEO	**Staff**	Hire staff. Each costs a salary and gives one benefit.
	Sabotage FROM ROUND 2	Hit a rival firm's cash, at a cost. A Legal Counsel blocks it.
Sales & Marketing	**Marketing**	Book one advert campaign to lift the next sales.
Finance	**Capital & Loans**	Borrow cash. The firm pays it back with interest.
	Investments	Buy property or funds. The value can rise or fall.
Operations	**Buying**	Hire the Procurement Specialist or the Stock Controller for the round.

B2B deal · Rounds 2 and 3

One firm sells a lot of 3 cars to another firm (4 in Round 3). With an odd number of firms, two firms offer to sell and the lower price wins.

Topic tasks

Each firm gets its own figures. The method is the same for every firm.

Task 1

ROUND 1

A new valeting bay.

Total profit over its life?

WORK OUT THE NUMBER · EXAMPLE FIGURES

Cost to set up	**£45,000**
Returns, each year	**£14,500**
It lasts	**5 years**

METHOD

Total profit = years × yearly returns − set-up cost

5 × £14,500 − £45,000 = £27,500

Why: It earns back its cost and £27,500 more.

Task 2

ROUND 1

You compare two investments by ARR.

Which two things does ARR not show?

PICK TWO

- ○ Profit as a % of the cost
- ○ The average profit a year
- ○ The cost of the investment
- ✔ **The risk that returns fall short**
- ✔ **When the money comes back**

Why: ARR averages the profit; it says nothing of timing or risk.

Task 3

ROUND 2

You work out the ARR of a new ramp.

In what order do you work it out?

PUT THESE IN ORDER, FIRST TO LAST · RIGHT ORDER

1. **Add up all the returns**
2. **Take off the cost: total profit**
3. **Divide by the years**
4. **Divide by the cost, × 100**

Why: Total profit, then profit a year, then as a % of the cost.

Task 4

ROUND 2

A new valeting bay.

Average rate of return (ARR), as a %?

WORK OUT THE NUMBER, THEN PICK IT · EXAMPLE FIGURES

Cost to set up	**£60,000**
Profit over its life	**£84,000**
It lasts	**5 years**

METHOD

ARR = profit ÷ years ÷ cost × 100

£84,000 ÷ 5 = £16,800; ÷ £60,000 × 100 = 28%

Why: It earns 28p a year for every £1 put in.

Task 5

ROUND 3

A ramp would have an ARR of 6%. You would borrow the money at 8% a year.

Should you invest, and why?

PICK A OR B AND GIVE A REASON

- ✔ **No**
- ○ Yes

Why: A 6% return does not cover an 8% cost of borrowing.

Task 6

ROUND 3

Which one pays better?

ARR of the better option, as a %?

WORK OUT THE NUMBER · EXAMPLE FIGURES

Option A: cost	**£75,000**
Option A: profit each year	**£9,000**
Option B: cost	**£55,000**
Option B: profit each year	**£13,750**

METHOD

Each option: ARR = profit a year ÷ cost × 100; take the better

A 12%; B 25%; better 25%

Why: Compare the rates, not the pounds of profit.

Final questions

6
questions for each pupil

150
Level 2 Business questions

4
of them on financial calculations: arr

The game draws the 4 questions on this topic, and 2 more from the 150 Level 2 questions.

1 A workshop upgrade costs £60,000 and is forecast to make total profit of £15,000 over three years. What is its average rate of return?
8.3%.

2 A project is forecast to make £3,000, £5,000 and £4,000 profit over three years. What maximum investment gives a 20% average rate of return?
£20,000.

3 Upgrade A: cost £50,000, £12,000 profit over three years. Upgrade B: cost £40,000, £10,800 over three years. Which has the higher average rate of return, and by how much?
Upgrade B, by 1 percentage point.

4 Two investments have similar average rate of return. What additional factor should the Finance lead consider?
Risk and when the cash comes back.